Nebula Teknoloji A.Ş.

Postmortem: INC-2026-0417 Primary Database Outage

Blameless incident review — NebulaCRM production (IST-1)

Document No	SRE-PM-2026-004
Severity	Sev-1
Incident commander	Ece Polat, DevOps Engineer
Author	Ece Polat
Reviewed by	Emre Kaya (IT Manager), Ayşe Demir (R&D Manager)
Review meeting	24 April 2026
Status	Final (action items updated 20 May 2026)
Issue Date	2026-05-20
Classification	Internal

1. Summary

On 17 April 2026, from 09:12 to 11:59 TRT (167 minutes), the primary PostgreSQL cluster in IST-1 stopped accepting writes. WAL archiving had stalled the previous evening after an object-storage credential rotation; un-archived WAL segments accumulated until the primary's data volume filled up, and the cluster manager fenced the primary into read-only mode to protect data integrity. During the incident 38% of API requests failed with 5xx errors and 286 customer accounts were affected. No customer data was lost.

This was our most serious production incident since INC-2025-1107 on 7 November 2025, a 52-minute Sev-2 outage caused by a DNS change on the load balancer. Unlike that incident, the trigger here was a routine change whose side effect was invisible to our monitoring for almost 15 hours.

2. Impact

- Duration: 167 minutes of degraded service (09:12–11:59 TRT).
- 38% of API requests failed; web application writes (creating or updating records) failed for all tenants on the shared cluster, while most reads continued to work from replicas.
- 286 customer accounts were affected. Tenants on dedicated Kurumsal Plus clusters were not affected.
- Outbound webhooks were delayed by up to 3 hours 10 minutes and were delivered through the normal retry schedule; no webhook was dropped.
- Monthly uptime for April 2026 was 99.61% for the shared production cluster.
- 61 Kurumsal and Kurumsal Plus customers were eligible for service credits totalling TRY 1.24M. Credits were calculated according to the service-credit tiers in the 2026 support SLA matrix.

3. Timeline (TRT)

Time	Event
16 Apr 18:30	Backup bucket credentials rotated (change CHG-2026-0388)
16 Apr 18:31	archive_command starts failing; failures retried silently
17 Apr 08:47	Primary disk passes 90% warning; alert goes to low-priority channel
17 Apr 09:12	Data volume full; primary fenced read-only; write errors begin
17 Apr 09:16	On-call paged by API 5xx alert

Time	Event
17 Apr 09:21	Sev-1 declared
17 Apr 09:24	Ece Polat takes the incident commander role
17 Apr 09:31	First status page update published
17 Apr 09:58	Root cause identified: WAL archiving stalled
17 Apr 10:20	Archiver restored with new credentials; backlog draining
17 Apr 11:05	Disk usage below 80%; writes re-enabled on primary
17 Apr 11:59	Error rates back to baseline; incident mitigated
17 Apr 14:30	Incident resolved on status page; customer summary sent

4. Root Cause

The backup bucket's access keys were rotated at 18:30 on 16 April as part of the quarterly key rotation. The rotation runbook updated the credentials used by the backup jobs but not the separate credential file read by PostgreSQL's archive_command on the primary. From 18:31 every archive attempt failed with an authorisation error. PostgreSQL correctly kept the un-archived WAL segments on local disk, so no data was at risk, but the volume grew by roughly 4% per hour.

Three conditions turned this into an outage. First, archive_command failures were not exported as a metric, so nothing alerted on growing archive lag. Second, the disk usage alert fired at 90% at 08:47 but was routed to a non-paging channel. Third, the old and new keys were not valid in parallel, so the rotation broke any consumer that had not been updated in the same minute.

5. What Went Well

- Detection of customer impact took 4 minutes from the first write error to the page.
- Replicas kept serving reads, so most dashboards and searches stayed available.
- Customer communication followed the Sev-1 template; the first status update went out 19 minutes after impact began.

6. Detection and Response Analysis

Time to detect customer impact was good, but time to detect the underlying fault was poor: the archiver had been failing for 14 hours and 41 minutes before the first write error. Responders spent the first 34 minutes treating the incident as a database capacity problem and attempted to extend the volume online, which bought about 20 minutes of writes before the volume filled again. The diagnosis changed at 09:58, when Murat Şen compared the archive status view with the replication lag dashboard and noticed that the last successfully archived segment dated from 18:30 the previous day.

We deliberately did not fail over to a replica. Promoting a replica would not have cleared the WAL backlog on the old primary and would have risked a split-brain during the credential fix. In hindsight this was the right call: once the archiver had valid credentials, the backlog of roughly 1,900 segments drained in 45 minutes and writes were re-enabled without a topology change.

7. Customer Communication

The status page at status.nebulacrm.example was updated six times during the incident. Account managers personally called the 12 largest affected customers, including two Kurumsal Plus accounts that asked for confirmation that their dedicated clusters were unaffected. A written summary of this postmortem was sent to all affected customer administrators on 30 April 2026. The support team handled 412 incident-related tickets on 17 and 18 April, about four times the normal daily volume.

8. Action Items

ID	Action	Owner	Due	Status
AI-1	Page on WAL archive lag > 10 min	Murat Şen	15 May 2026	Done
AI-2	Disk alert 90% → 75%, paging	Ece Polat	30 Apr 2026	Done
AI-3	Dual-key credential rotation runbook	Emre Kaya	31 May 2026	Done
AI-4	Quarterly database failover game day	Ece Polat	30 Jun 2026	In progress
AI-5	Export archive_command failures as metric	Okan Yıldız	15 May 2026	Done
AI-6	Sev-1 customer notice within 30 min	Can Öztürk	15 May 2026	Done

9. Lessons Learned

Routine changes deserve the same verification as feature releases: a credential rotation is only complete when every consumer has been confirmed on the new key. We also learned that a warning nobody is paged for is effectively not an alert. The dual-key rule from AI-3 has since been added to the IT password and access policy, and the backup and disaster recovery plan now lists the WAL lag and disk thresholds as mandatory monitoring.